AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 13 · STUDENT LESSON

Types of Classification Error

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 12: accuracy counts correct predictions over all scored cases.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Build a labelled confusion matrix from matched reference and prediction pairs. Try the recall before opening notes.

Your learning target

I can build a labelled **confusion matrix** from matched reference and **prediction** pairs.

I can distinguish **false positive**s from **false negative**s using a declared **positive class**.

I can check matrix totals and explain why equal accuracy can hide different errors.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Accuracy as a Fraction. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 13: Types of Classification Error

False alarms and misses can have different costs.



THINKER MISSION

Sort 12 **prediction**s into four outcome boxes.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

> Why should a reproducible run save **prediction**s linked to example IDs rather than only an accuracy score?

This week's end goal

By the end, I can construct a **confusion matrix** with declared axes and name both error types.

STUDENT LESSON · 2 OF 18

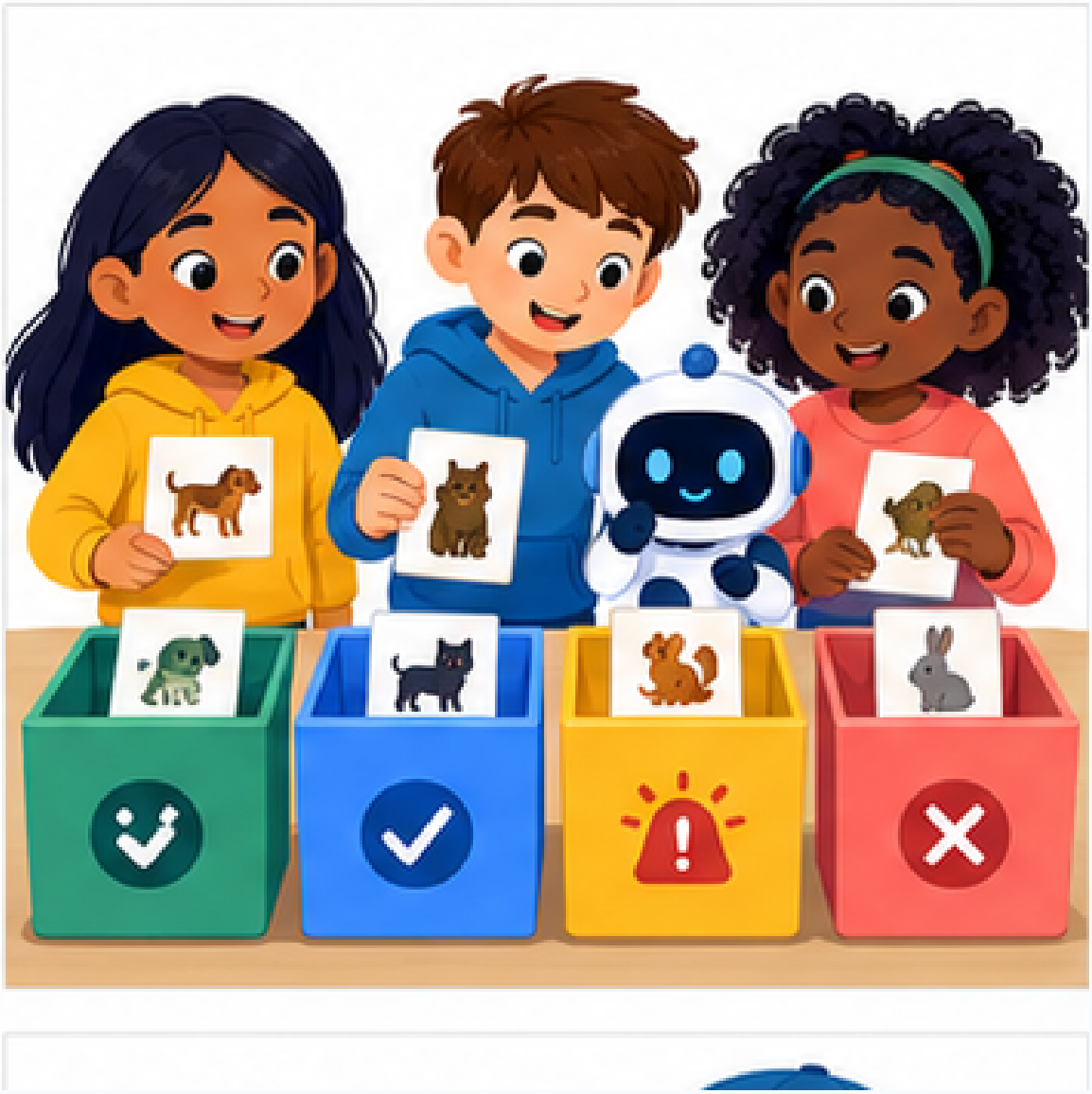
Notice the evidence

> Investigate the supplied Types of Classification Error case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use types of classification error to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

A **confusion matrix** shows which labels a classifier mixes up. It counts examples after **prediction**s are made; it does not fit or repair the classifier.

Week 12 had reference/**prediction** pairs clear/clear, blocked/blocked, clear/blocked. A labelled table can preserve the type of that error as well as the total score.

This week uses eight new fictional evaluation records from a frozen classifier. They are not extra **prediction**s from the Week 12 code. Each ID appears once with a reviewed reference and one **prediction**.

The grouped table below lists every ID. Keep reference and **prediction** paired by ID before counting.

Evaluation IDs	Reference	Prediction
V1 V2	clear	clear
V3	clear	blocked
V4 V5	blocked	clear

Evaluation IDs	Reference	Prediction
V6 V7 V8	blocked	blocked

Use reference labels for rows and predicted labels for columns. On both axes the order is clear, then blocked. Declare blocked as the **positive class** and clear as negative.

Positive means the condition of interest, not good or safe. A missed blocked case and a false alarm are different errors.

STUDENT LESSON · 3 OF 18

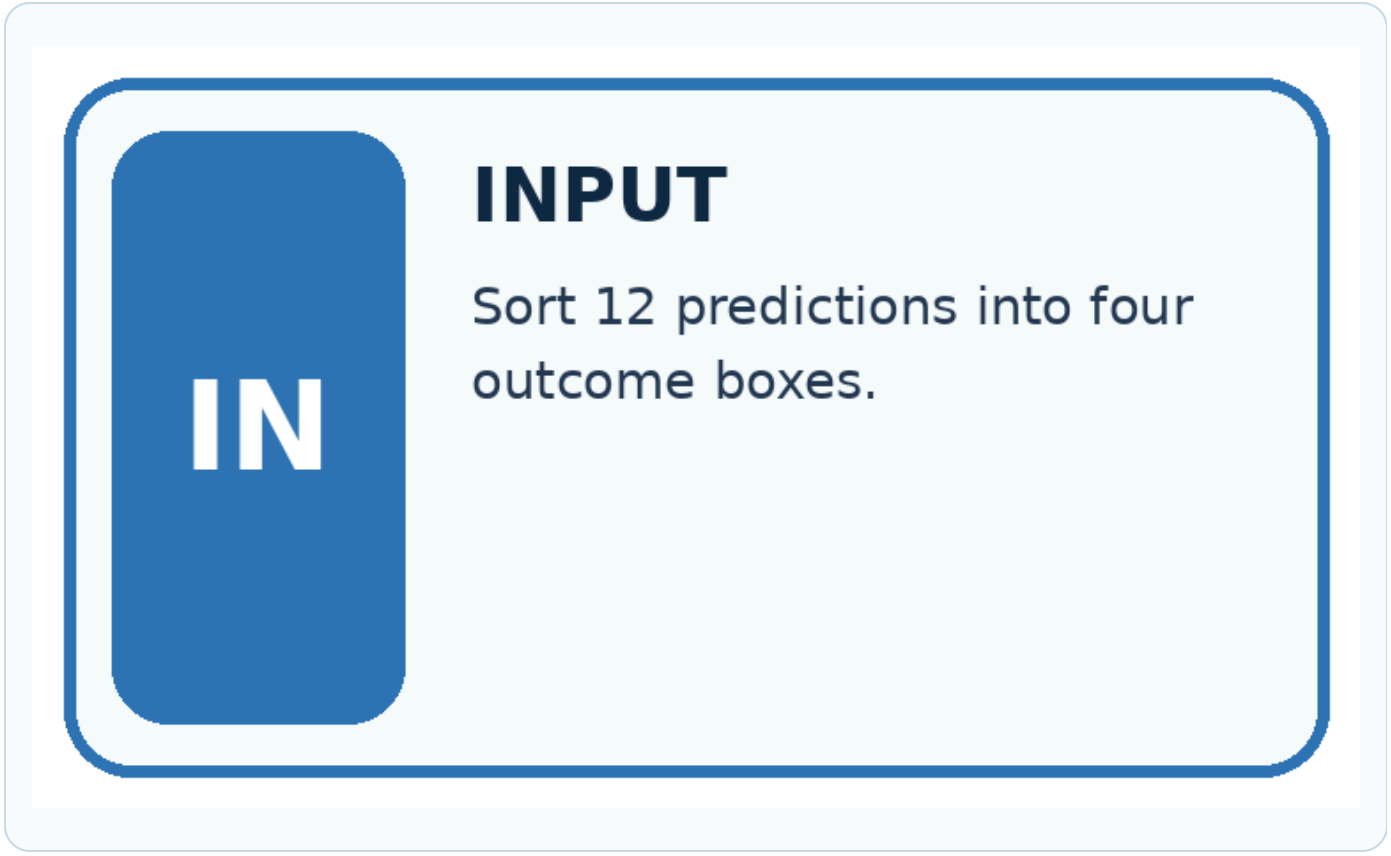
See how it works

> Follow the mechanism for Types of Classification Error. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → **Process** → **Output**

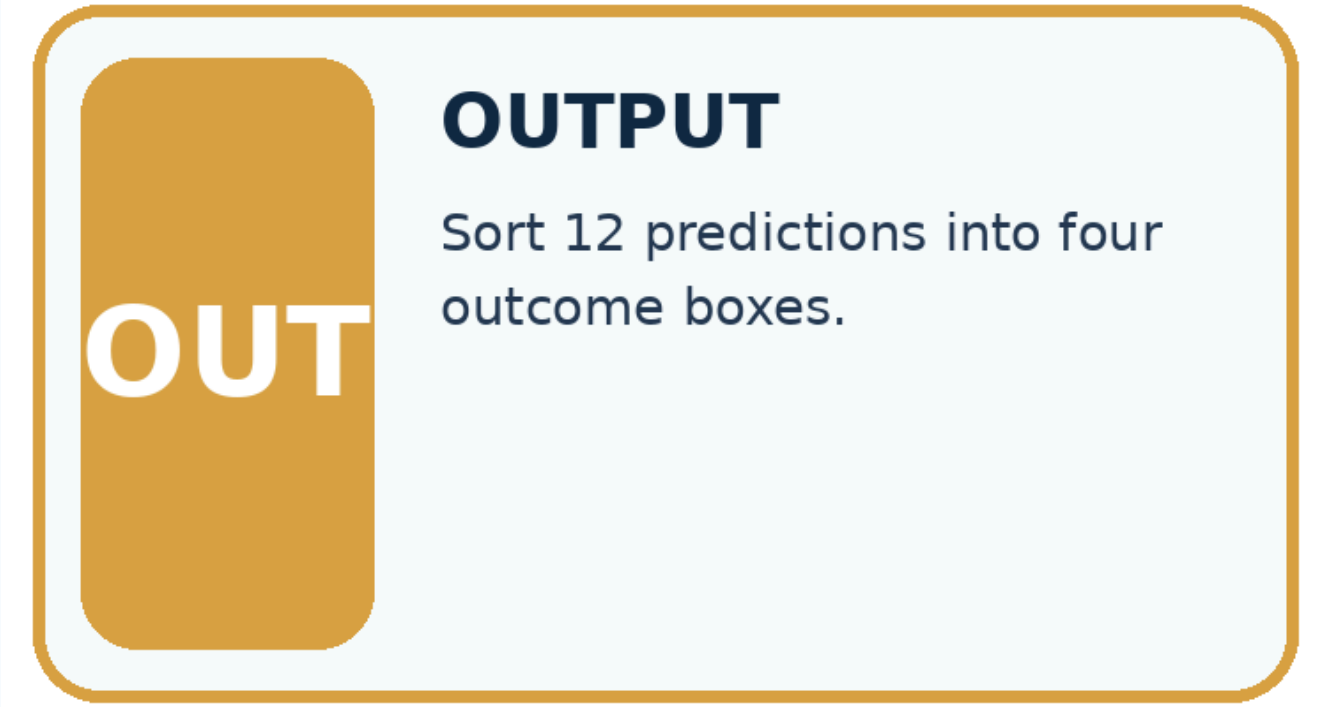
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Sort 12 predictions into four outcome boxes.

↓


PROCESS
False alarms and misses can have different costs.

↓



Which stage could fail even when the other two are correct? Explain:

Write or draw your reasoning and evidence.

A closer explanation

```
pairs = [("V1", "clear", "clear"),
         ("V2", "clear", "clear"),
         ("V3", "clear", "blocked"),
         ("V4", "blocked", "clear"),
         ("V5", "blocked", "clear"),
         ("V6", "blocked", "blocked"),
         ("V7", "blocked", "blocked"),
         ("V8", "blocked", "blocked")]
```

```
labels = ["clear", "blocked"]
index = {label: i for i, label in enumerate(labels)}
counts = [[0, 0], [0, 0]]
for item_id, reference, prediction in pairs:
    counts[index[reference]][index[prediction]] += 1
print(counts)
correct = counts[0][0] + counts[1][1]
print(correct, len(pairs))
```

index maps clear to 0 and blocked to 1. Each paired record adds one to its reference row and **prediction** column. The script prints [[2, 1], [2, 3]] then 5 8. It assumes unique IDs and only these two labels.

Check IDs and missing or unresolved references before counting; the code does not validate all those conditions.

Reference rows	Predicted clear	Predicted blocked
Reference clear	2	1
Reference blocked	2	3

The diagonal cells count correct **prediction**s: 2+3=5. Off-diagonal cells count mistakes: 1+2=3. Every record must contribute exactly once. This is a count matrix, not percentages or probabilities.

Keep the axis labels even when copying the numbers.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

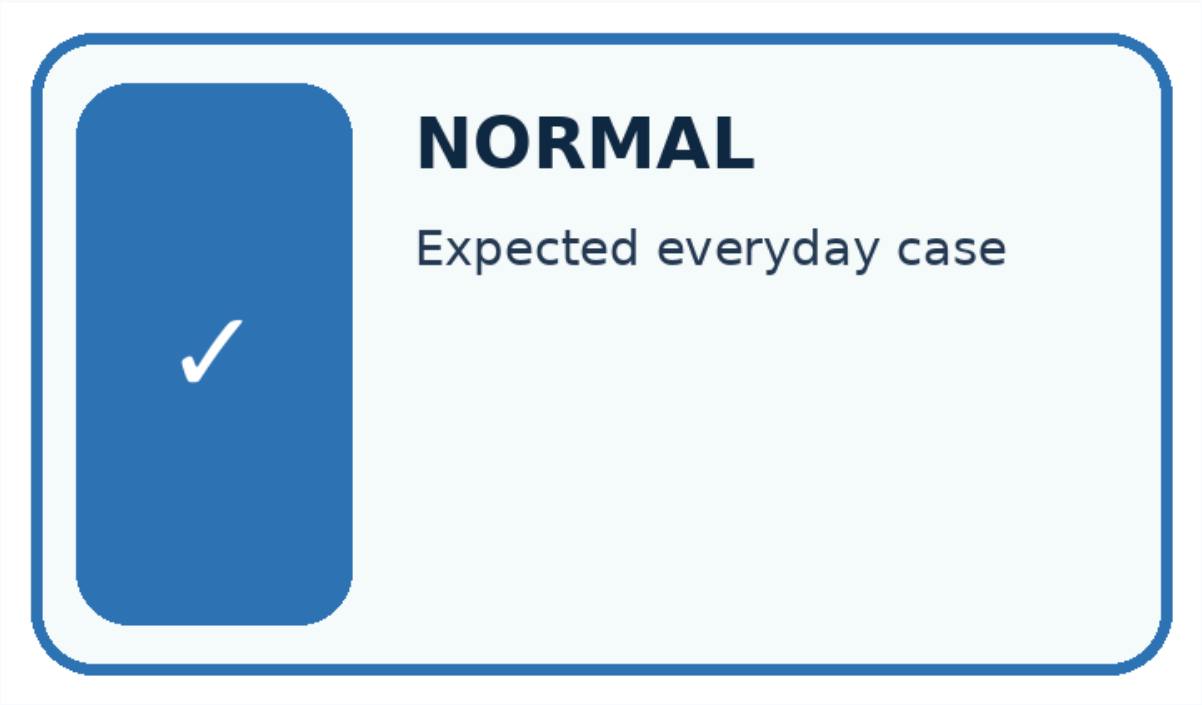
> Compare the supplied cases for Types of Classification Error. Identify the detail that changes the conclusion.

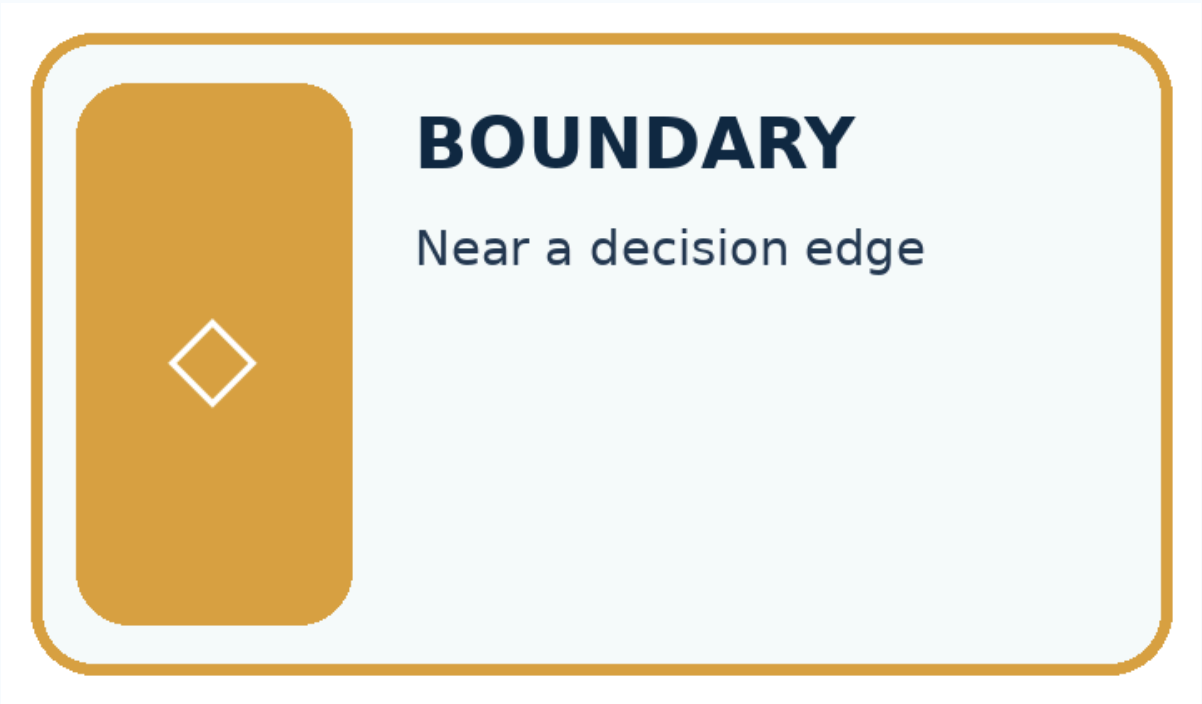
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

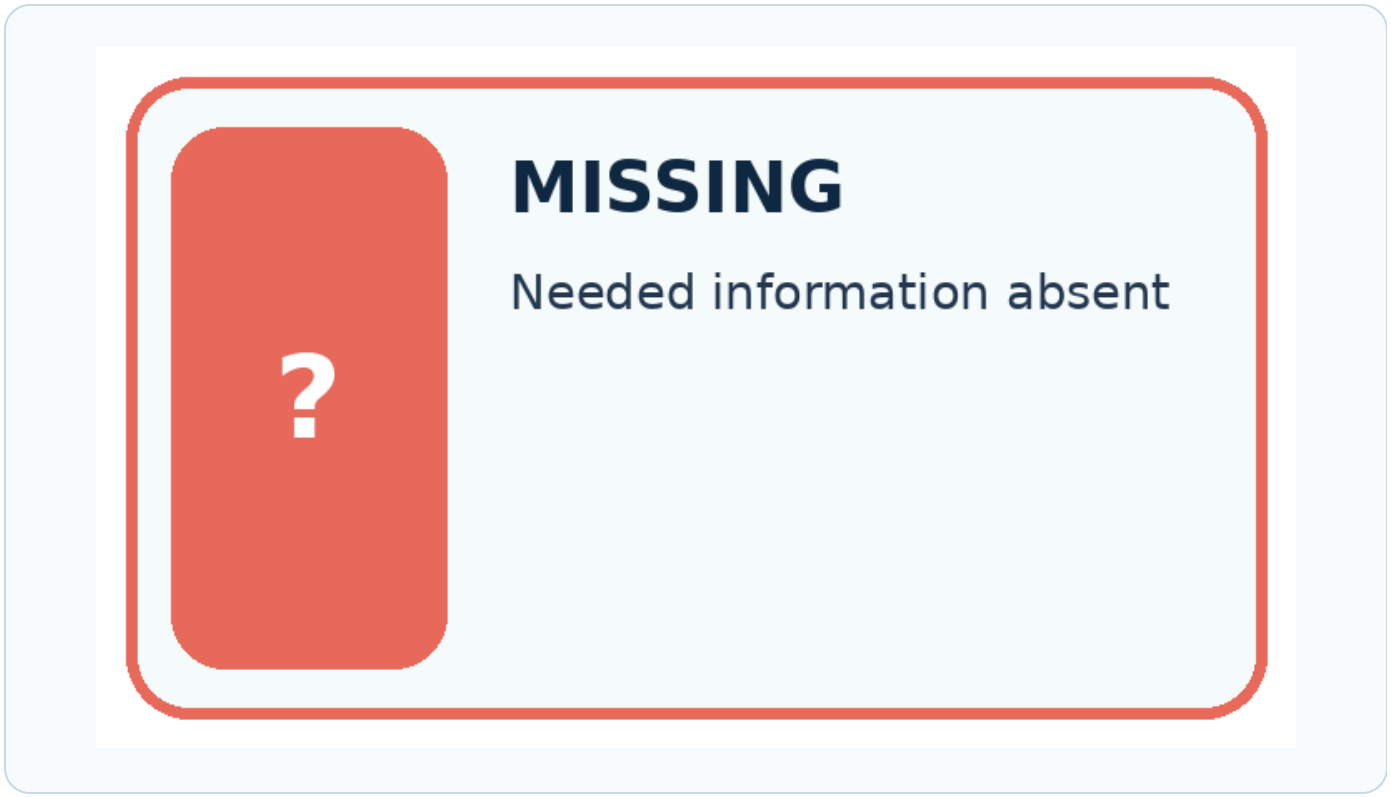
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: Change one repair/sound **prediction** to repair/repair after a real model revision.

Predicted result

Prediction: Repair row becomes [0,3]; agreement becomes 5/6.

Why the result follows

Reason: One **false negative** moves to **true positive**; the sound row stays [2,1].

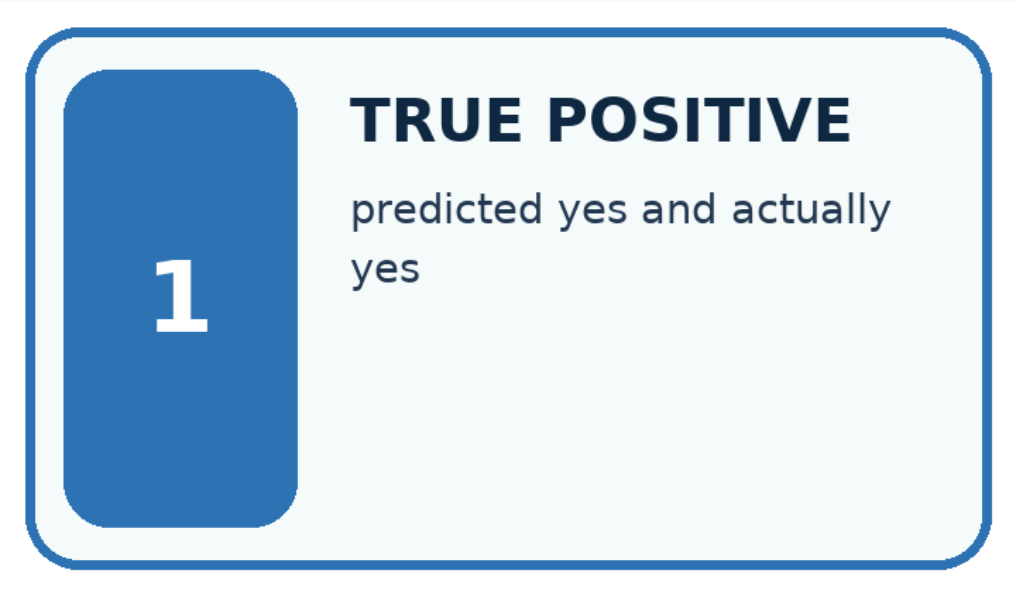
Words that unlock the idea

> Use the vocabulary of Types of Classification Error to explain a concrete decision, rather than only memorising definitions.

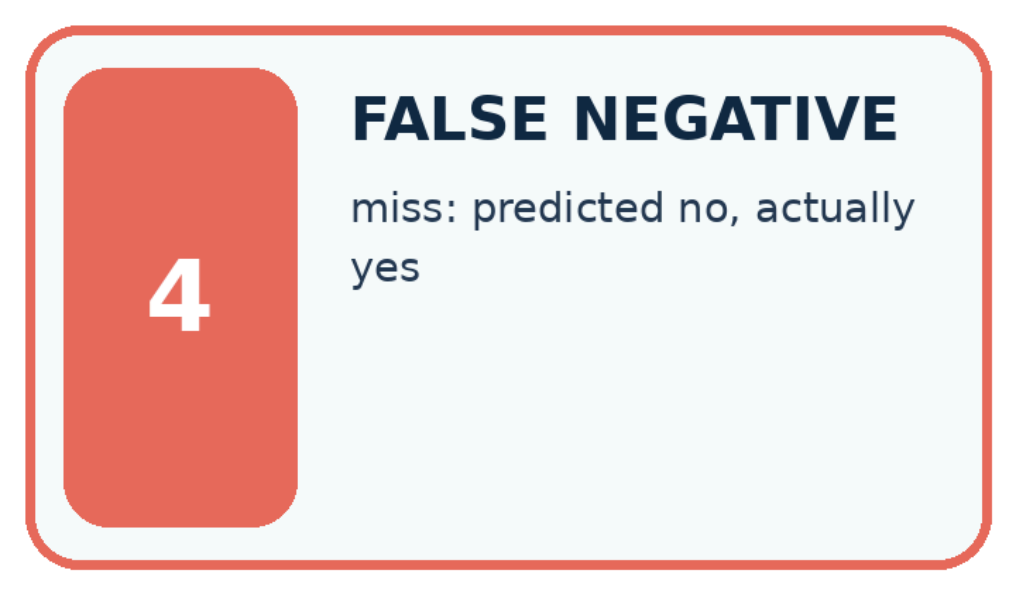
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
TRUE NEGATIVE
predicted no and actually no
3
FALSE POSITIVE
false alarm: predicted yes, actually no



Word	My own example	Not this / boundary
true positive	An actually positive case correctly predicted positive.	
true negative	An actually negative case correctly predicted negative.	
false positive	An actually negative case predicted positive, under a stated definition of positive.	
false negative	An actually positive case predicted negative, under a stated definition of positive.	

Words used in the closer explanation

Word	Meaning
Confusion matrix	A table counting reference and prediction combinations
Positive class	The class chosen as the condition of interest

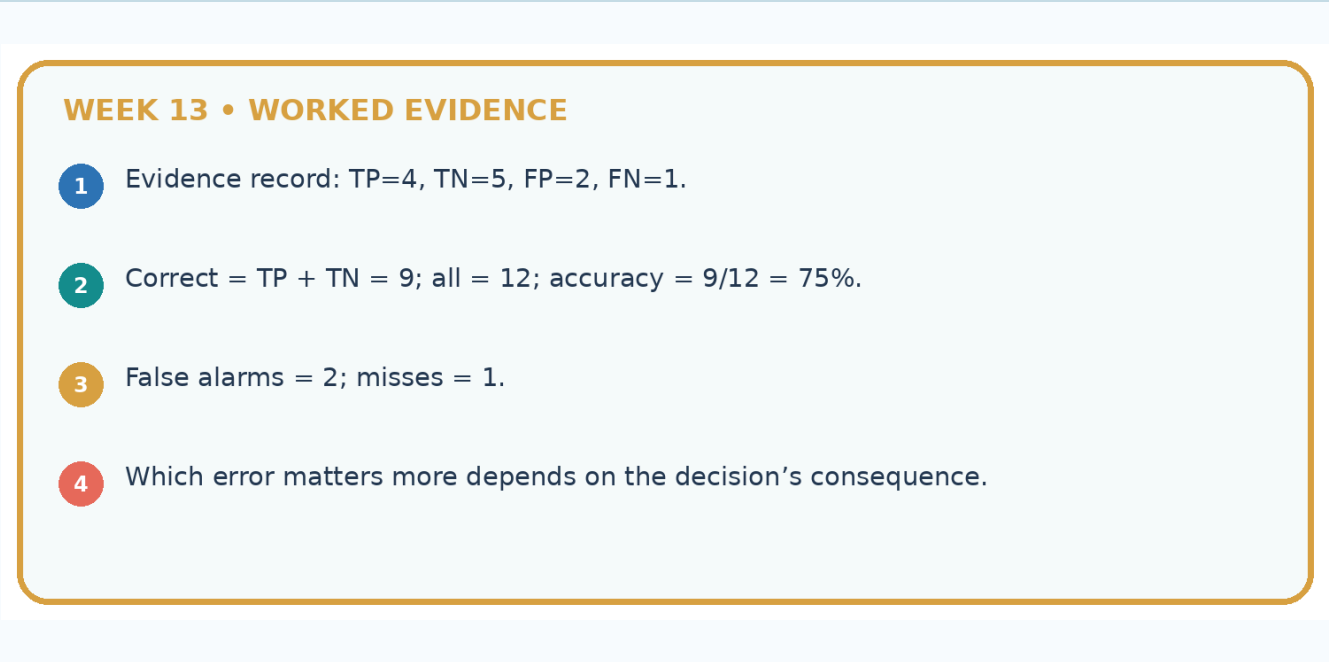
Follow a worked example

> Follow the worked trace for Types of Classification Error, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. **Evidence** record: TP=4, TN=5, FP=2, FN=1.

2. Correct = TP + TN = 9; all = 12; accuracy = 9/12 = 75%.

3. False alarms = 2; misses = 1.

4. Which error matters more depends on the decision's consequence.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Blocked is the positive class

REFERENCE AND PREDICTION	CELL NAME	IDS AND COUNT
clear then clear	True negative TN	V1 V2 count 2
clear then blocked	False positive FP	V3 count 1
blocked then clear	False negative FN	V4 V5 count 2
blocked then blocked	True positive TP	V6 V7 V8 count 3

Read reference first and prediction second. Here a false positive is a false blocked alarm; a false negative is a missed blocked case.

Place V3 and V4 in the matrix. Explain which is a false positive and which is a false negative when blocked is positive.

True means **prediction** agrees with the reference. It does not prove that the reference annotation is infallible. Resolve disputed labels with the documented review policy rather than silently changing them to match **prediction**s.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Types of Classification Error

A fictional book-condition checker predicts repair or sound.

The facts we will use

Supplied facts

Use reference rows and **prediction** columns in order sound,repair. Repair is positive.

Supplied facts

Six reference/**prediction** pairs: sound/sound twice; sound/repair once; repair/sound once; repair/repair twice.

AI ACADEMY / AI-2/13

Reference rows, prediction columns

	Predicted sound	Predicted repair
Reference sound	2	1
Reference repair	1	2

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

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Types of Classification Error

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Declare axes

Rows: reference sound,repair. Columns: predicted sound,repair.

2. Fill sound row

Two true negatives and one false positive: [2,1].

3. Fill repair row

One false negative and two true positives: [1,2].

4. Read result

Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.

CHANGE ONE THING

Change one repair/sound prediction to repair/repair after a real model revision.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Declare axes	Rows: reference sound,repair. Columns: predicted sound,repair.
2. Fill sound row	Two true negatives and one false positive: [2,1].
3. Fill repair row	One false negative and two true positives: [1,2].
4. Read result	Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Each case enters one cell according to its reference and predicted label. The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair.

Trace the cell definitions before naming rates; swapping axes changes the table's interpretation even if its diagonal sum remains unchanged.

What this example does not establish

Keep the stated limit

Reference labels can themselves be disputed; a true-positive name does not guarantee the annotation is infallible.

Keep the stated limit

Read the labelled picture: Which error causes a sound book to be sent for unnecessary repair review?

Keep the stated limit

Change one thing: Change one repair/sound **prediction** to repair/repair after a real model revision. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Types of Classification Error claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: All mistakes should be treated as one identical error type. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Always report the evaluated split, number of records, reference and **prediction** axes, label order and **positive class**. Reversing the axis order changes positions, not the underlying outcomes.

Transposing rows and columns swaps the two kinds of error unless the labels are corrected. Read the headers before interpreting a diagram from another source.

Check the claim against the facts

A zero cell means no such event in this evaluated set, not that it is impossible. A **confusion matrix** does not show the cause of errors, label quality or performance on future data. Keep the ID-level records for investigation.

Next week examines accuracy when class counts are unequal.

Two classifiers have accuracy 5/8. One misses two blocked records and the other misses three. Explain how both statements can be true and what must accompany their matrices for a fair comparison.

STUDENT LESSON · 8 OF 18

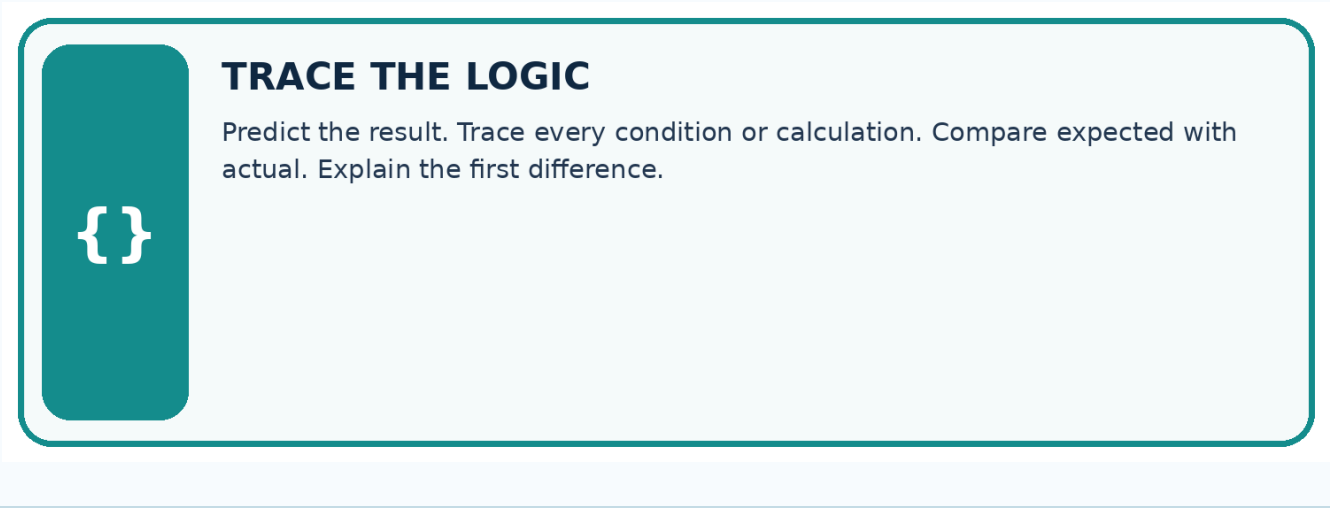
Practise together

Practise Types of Classification Error with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Row totals are 3 reference-clear and 5 reference-blocked. Column totals are 4 predicted-clear and 4 predicted-blocked. All four cells sum to 8. Accuracy is the diagonal total divided by the total: 5/8, or 62.5 percent.

The Week 12 three-record matrix was [[1,1],[0,1]] in this same axis order, giving 2/3.

Another classifier on these same eight reference-labelled records could have matrix [[3,0],[3,2]]. It also gets 5/8 correct, but has zero **false positive**s and three **false negative**s.

Our worked classifier has one **false positive** and two **false negative**s. Equal accuracy does not settle which error pattern is preferable; consider the task and **evidence** before choosing.

Practise with less help: Explain the two error types

Use workbook W2 for the connected example “Confusion matrices”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Confusion matrices. Proposed explanation: “Positive means good”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Which error causes a sound book to be sent for unnecessary repair review?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

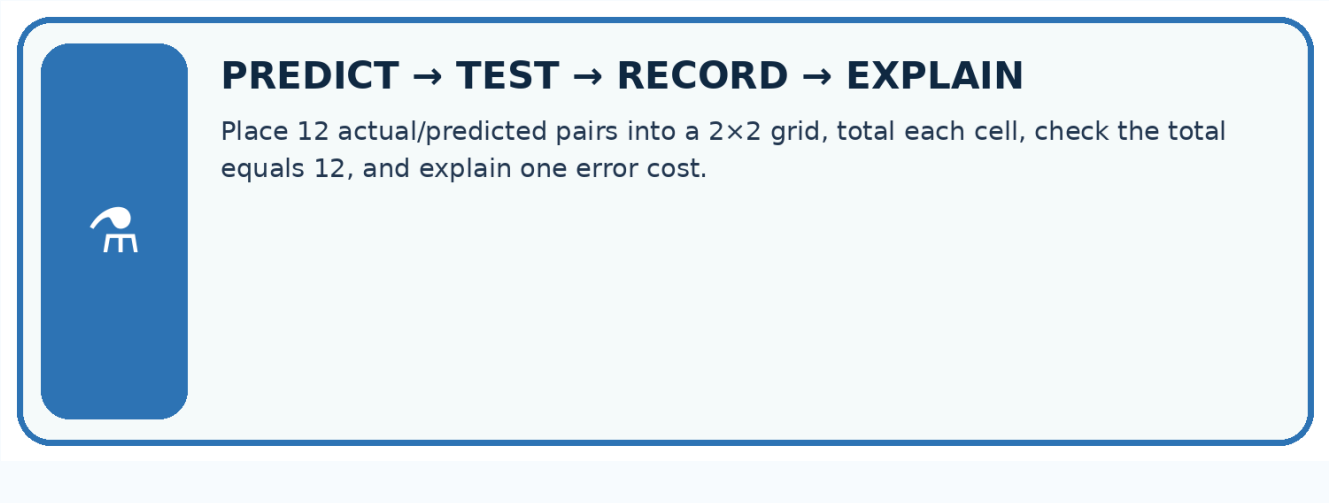
Run the investigation

> Investigate Types of Classification Error. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Place 12 actual/predicted pairs into a 2×2 grid, total each cell, check the total equals 12, and explain one error cost.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

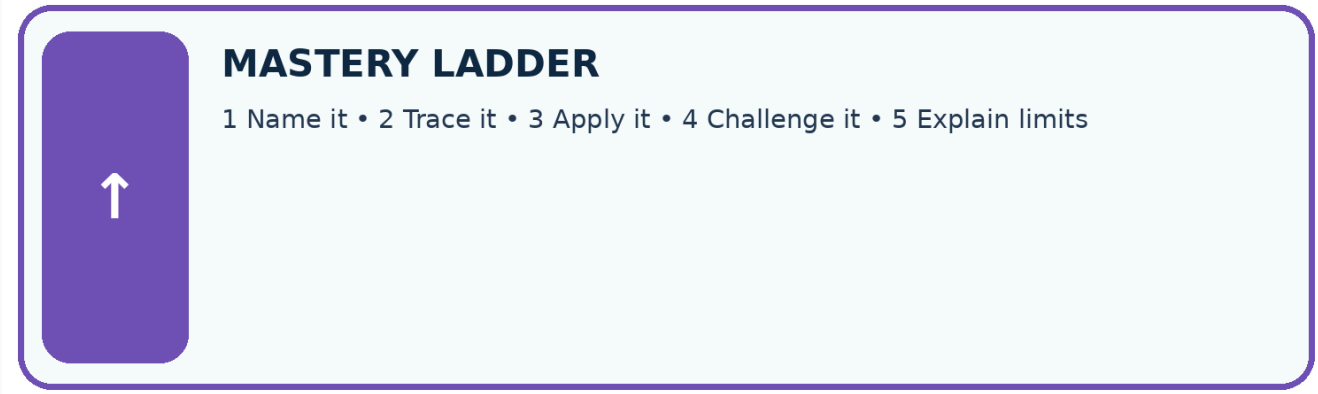
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Types of Classification Error. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

> **Write or draw your reasoning and evidence.**

STUDENT LESSON · 11 OF 18

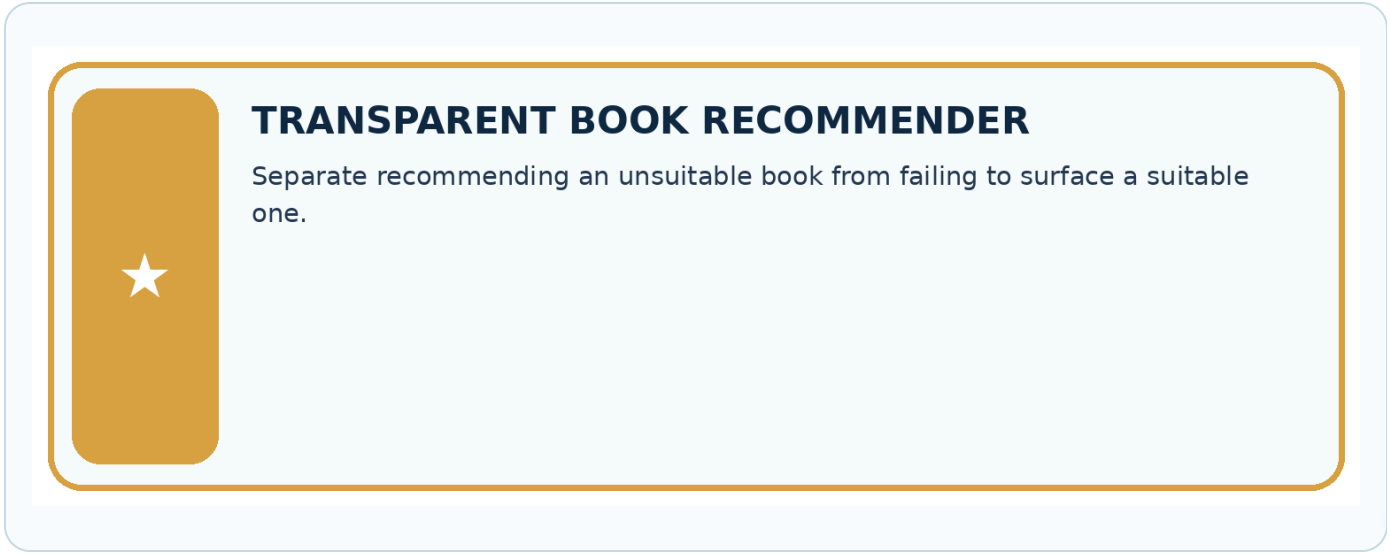
Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Separate recommending an unsuitable book from failing to surface a suitable one.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Use a confusion table when evaluating recommender category **prediction**s and attach concrete consequences to each error.

STUDENT LESSON · 12 OF 18

Show what you know

Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.

✓

INDEPENDENT PROOF

Sort 12 predictions into four outcome boxes.

Try a more demanding case

CHALLENGE Sort 12 **prediction**s into four outcome boxes.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Types of Classification Error case for supported practice. Repeated examples do not become fresh assessments.

Week 13 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Positive means should recommend. Actual/predicted pairs are yes/yes, no/yes, yes/no, no/no.

The question

Name the four outcomes.

Worked explanation

True positive, **false positive**, **false negative**, **true negative**. Define positive before naming errors.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Name outcomes using actual and predicted labels

Evidence or stage	Value or result
True positive	1
False negative	1
False positive	1
True negative	1

Positive means should recommend. Actual/predicted pairs are yes/yes, no/yes, yes/no, no/no.

Name the four outcomes.

True positive, **false positive**, **false negative**, **true negative**. Define positive before naming errors.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Count a fresh evaluation independently

Each pair below is reference then **prediction**: R1 clear/clear, R2 clear/blocked, R3 clear/blocked, R4 blocked/blocked, R5 blocked/clear, R6 blocked/blocked. Use clear then blocked on both axes and reference rows.

Build the matrix, give row and column totals, name TP,TN,FP,FN with blocked positive, and calculate accuracy.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 13 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Positive means sports. A sports book predicted other and a non-sports book predicted sports occur. Name each error and its consequence for the sports shelf.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Separate recommending an unsuitable book from failing to surface a suitable one.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

Positive means 'should recommend'. A book that should be recommended is rejected; a book that should not be recommended is accepted. Name both errors and explain who experiences each consequence.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Types of Classification Error, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional book-condition checker predicts repair or sound.

By the end, I can	How I will show it
Build a labelled confusion matrix from matched reference and prediction pairs	Explain the deciding step, show a trace or test, and state one limit.
Distinguish false positives from false negatives using a declared positive class	Explain the deciding step, show a trace or test, and state one limit.

By the end, I can	How I will show it
Check matrix totals and explain why equal accuracy can hide different errors	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Types of Classification Error without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Build a labelled **confusion matrix** from matched reference and **prediction** pairs

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional book-condition checker predicts repair or sound.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.

1. Declare axes

2. Read result

3. Fill repair row

4. Fill sound row

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Read result Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3. Each case enters one cell according to its reference and predicted label.

The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair.

Trace the cell definitions before naming rates; swapping axes changes the table's interpretation even if its diagonal sum remains unchanged.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Change one repair/sound **prediction** to repair/repair after a real model revision. Which conclusion is supported by the supplied facts?

1. Repair row becomes [0,3]; agreement becomes 5/6.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Repair row becomes [0,3]; agreement becomes 5/6. One **false negative** moves to **true positive**; the sound row stays [2,1].

Repair a missing building block

Review Accuracy as a Fraction (ai-2/12). Goal: Explain why overall accuracy can hide failure on a less common class. Short practice: Why is 100% repair detection not sufficient to declare the changed classifier best?.

Return to this week and explain what you changed.

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

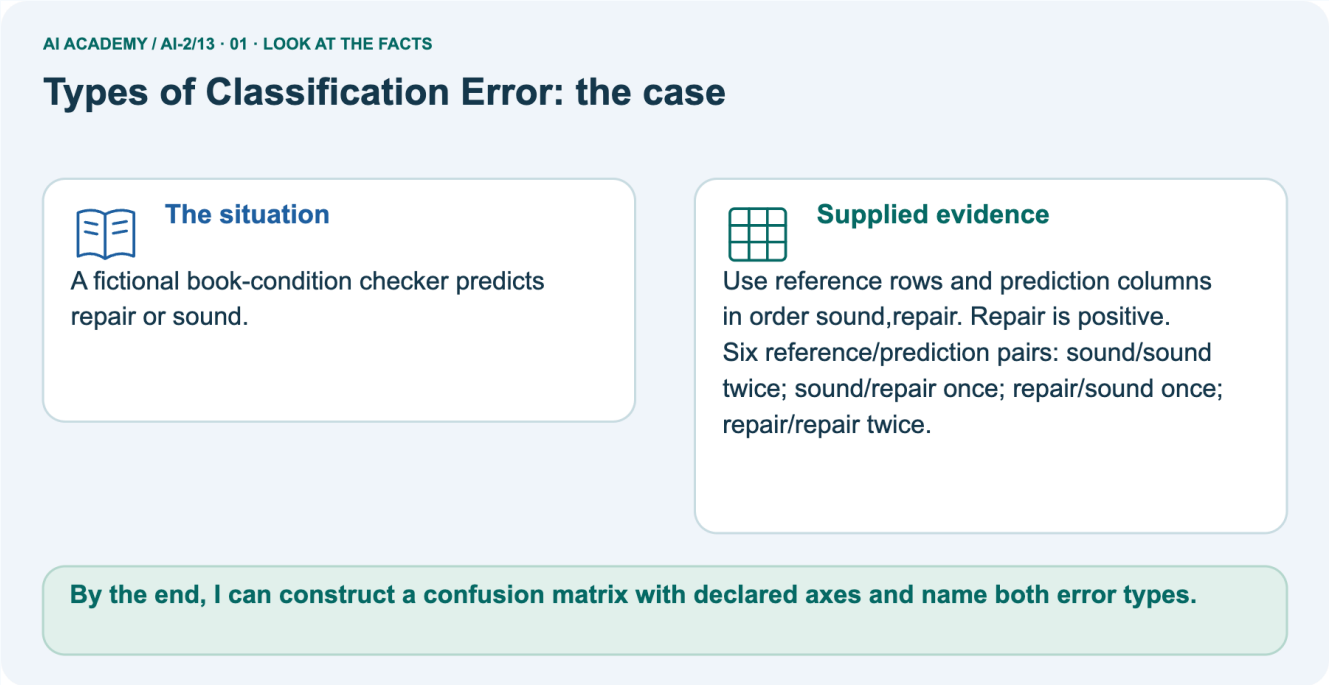
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Types of Classification Error graphic using its labels, deciding evidence and relevant limitation.

01. Types of Classification Error: the case



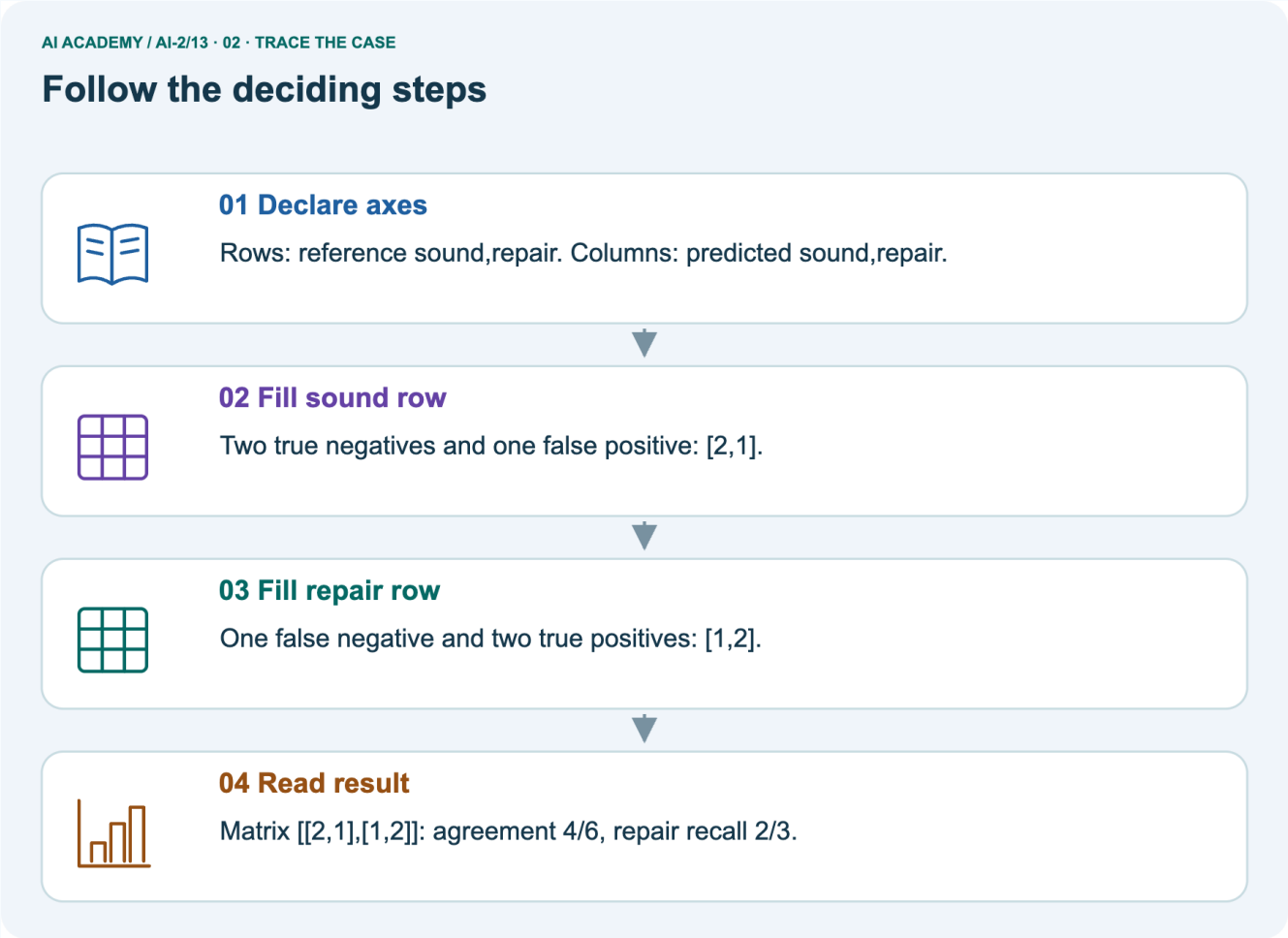
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional book-condition checker predicts repair or sound. Supplied facts: Use reference rows and **prediction** columns in order sound,repair. Repair is positive.

Six reference/**prediction** pairs: sound/sound twice; sound/repair once; repair/sound once; repair/repair twice.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



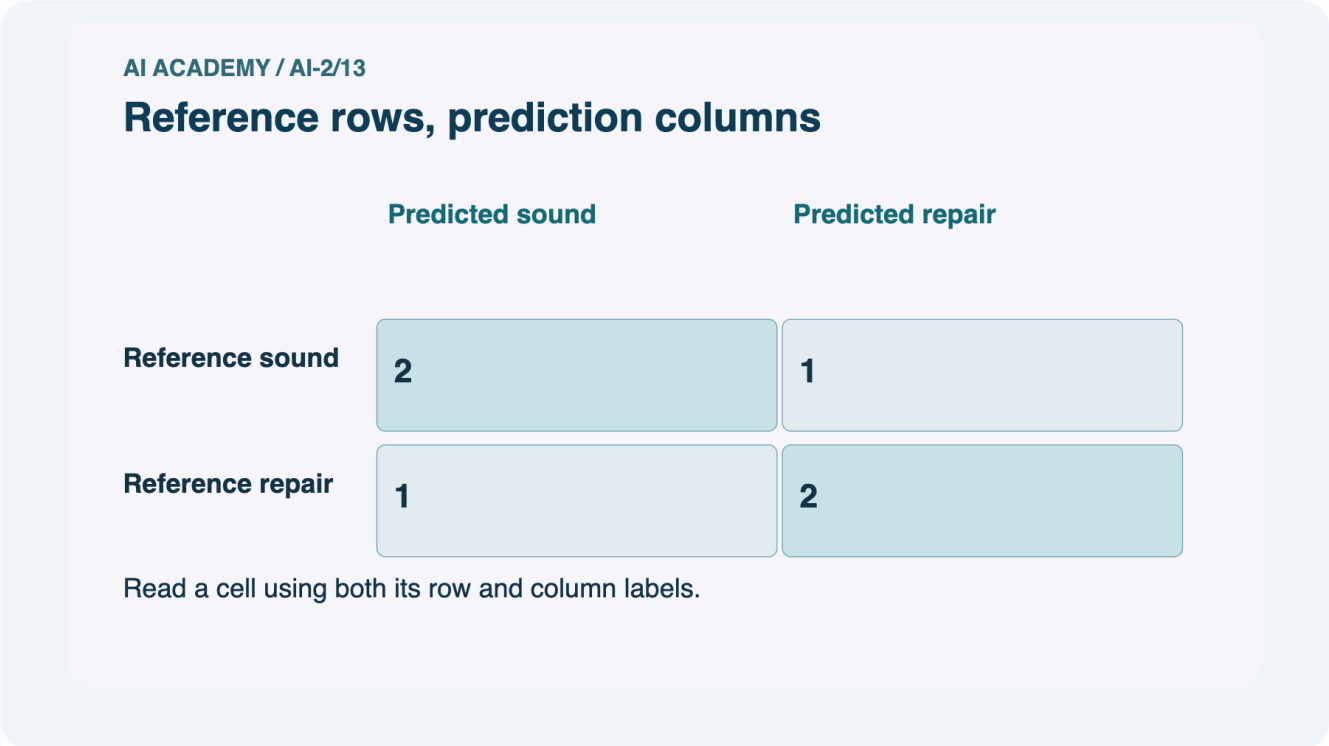
Follow the labelled picture with the matching explanation. Open the vector diagram

Declare axes: Rows: reference sound,repair.

Columns: predicted sound,repair. → Fill sound row: Two **true negative**s and one **false positive**: [2,1]. → Fill repair row: One **false negative** and two **true positive**s: [1,2]. → Read result: Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Reference rows, prediction columns

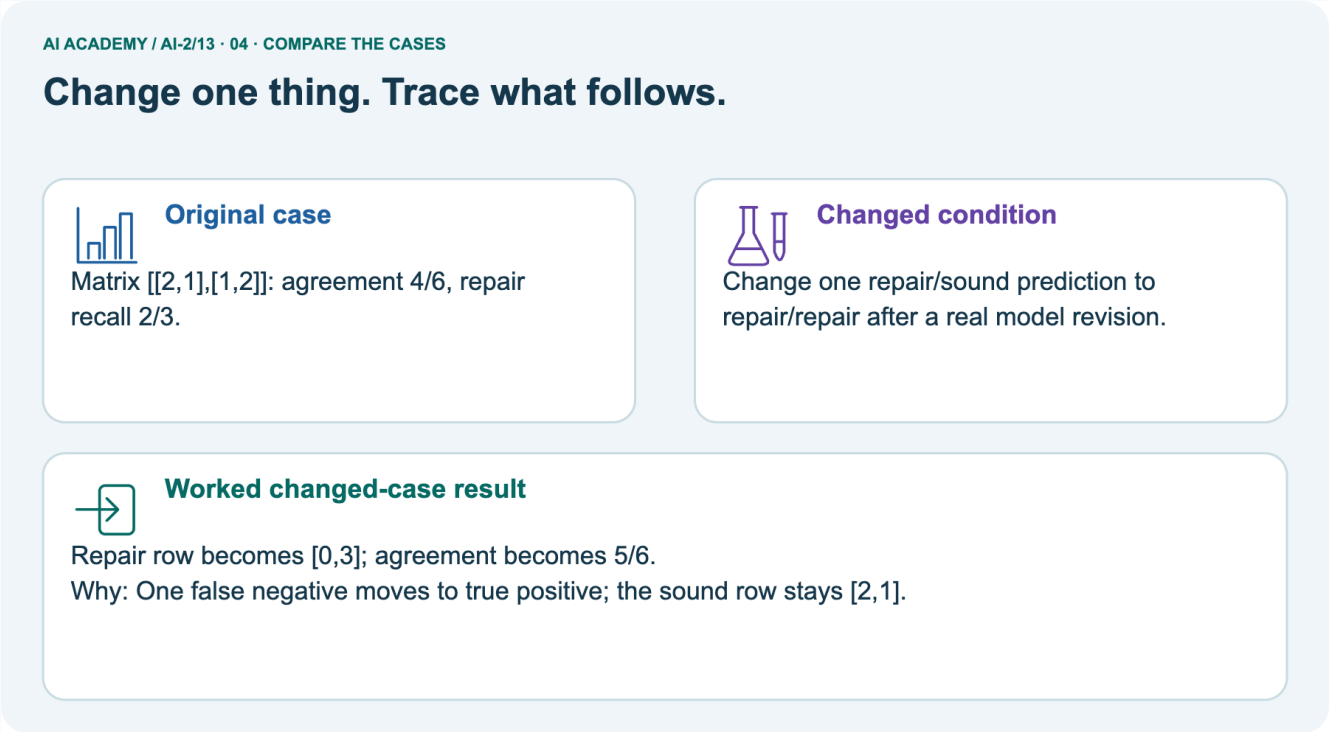


Follow the labelled picture with the matching explanation. Open the vector diagram

Reference rows, **prediction** columns. Reference sound: Predicted sound = 2, Predicted repair = 1. Reference repair: Predicted sound = 1, Predicted repair = 2.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



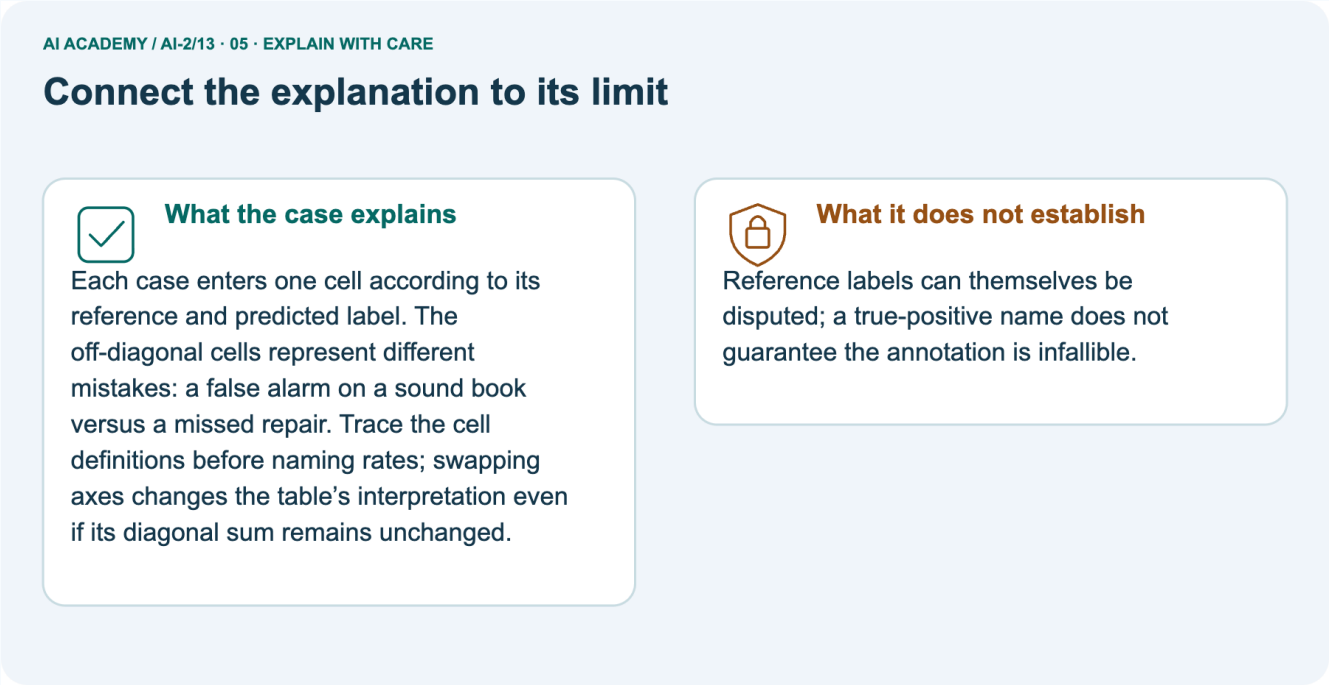
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Matrix [[2,1],[1,2]]: agreement 4/6, repair recall 2/3. Changed condition: Change one repair/sound **prediction** to repair/repair after a real model revision.

Expected result: Repair row becomes [0,3]; agreement becomes 5/6. Reason: One **false negative** moves to **true positive**; the sound row stays [2,1].

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

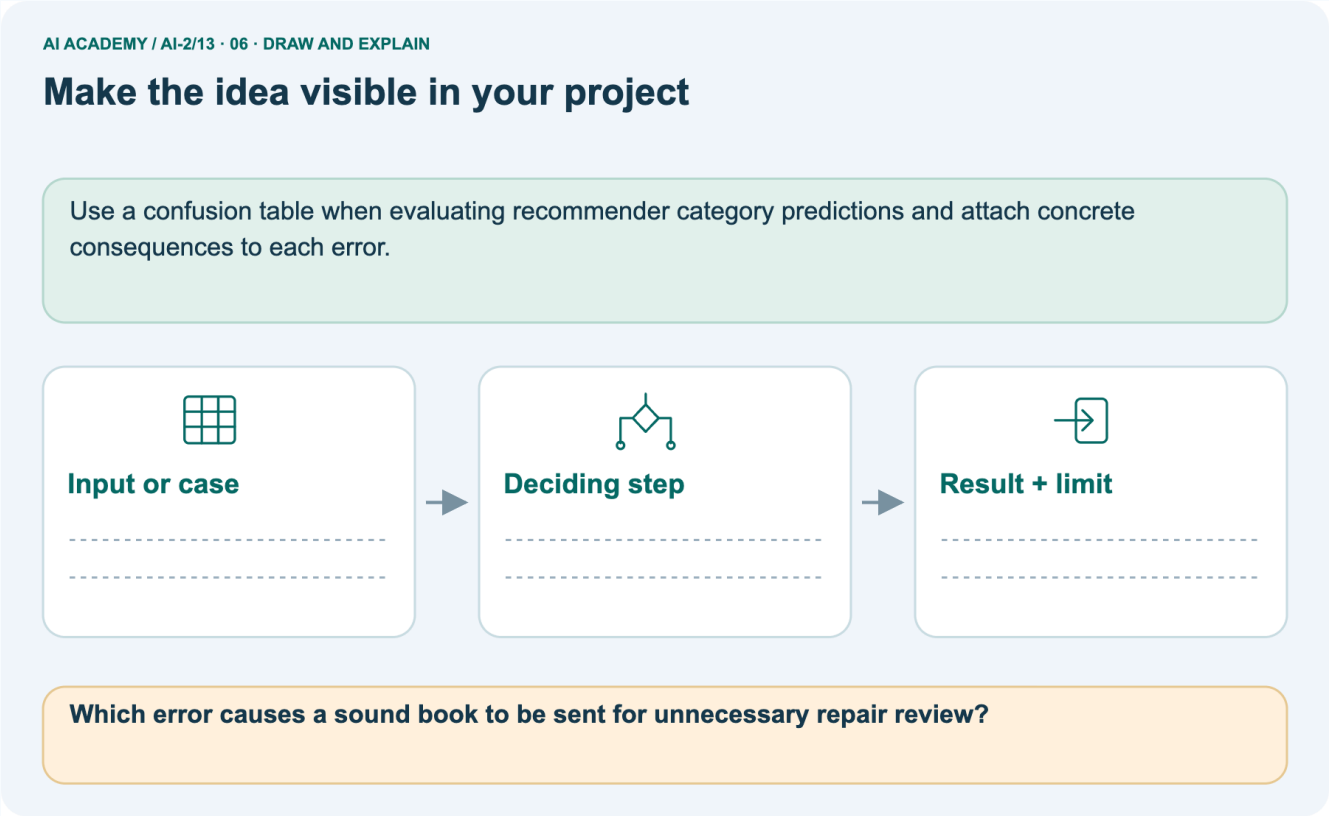
Each case enters one cell according to its reference and predicted label. The off-diagonal cells represent different mistakes: a false alarm on a sound book versus a missed repair.

Trace the cell definitions before naming rates; swapping axes changes the table’s interpretation even if its diagonal sum remains unchanged.

Limit: Reference labels can themselves be disputed; a true-positive name does not guarantee the annotation is infallible.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use a confusion table when evaluating recommender category **prediction**s and attach concrete consequences to each error. Draw your **input**, deciding step, result and limit.

Supported question: Which error causes a sound book to be sent for unnecessary repair review?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.